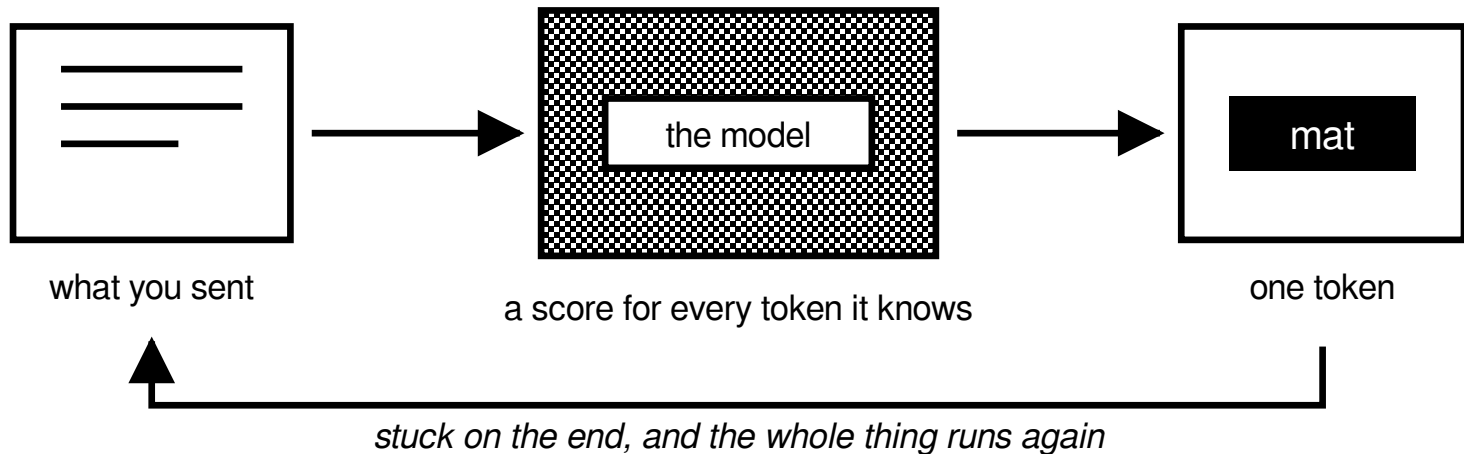


LLMs for n00bs

Nine slides on what a language model actually does, and what it cannot. No maths.

The last one asks a real one, if you have set one up. Press Next.



Back

Next

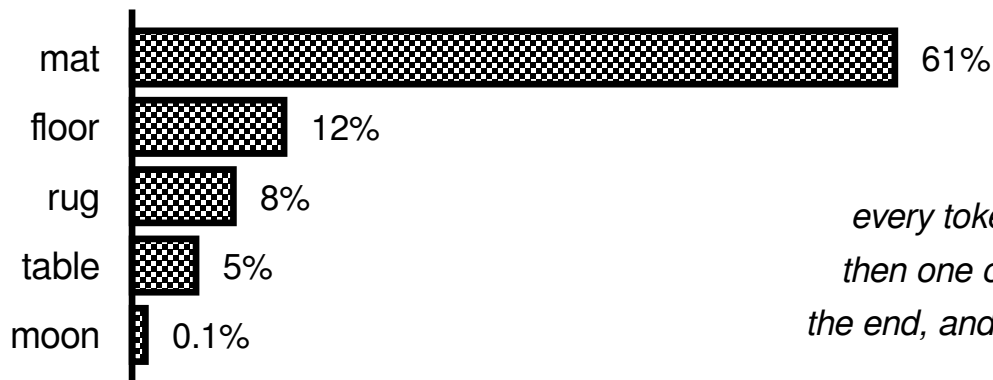
Start Over

It predicts the next token

Given some text, it scores every token that could come next, picks one, sticks it on the end, and starts again.

That is the whole mechanic. Everything on the slides after this one is a consequence of it.

The cat sat on the



*every token it knows gets a number.
then one of them is picked, added to
the end, and the question asked again.*

Back

Next

Start Over

A token is not a word

Text is chopped up before the model ever sees it. Common words survive whole; rarer ones become several pieces.

Which is why it is bad at questions about spelling. It was never shown the letters.

un	bel	iev	able
----	-----	-----	------

301

6789

12144

1122

four pieces, one word

str	aw	berry
-----	----	-------

three pieces, and not one of them is an "r"

Common words are one piece.

Rare ones are several.

Asking how many r's are in
"strawberry" asks about letters
it was never shown.

Back

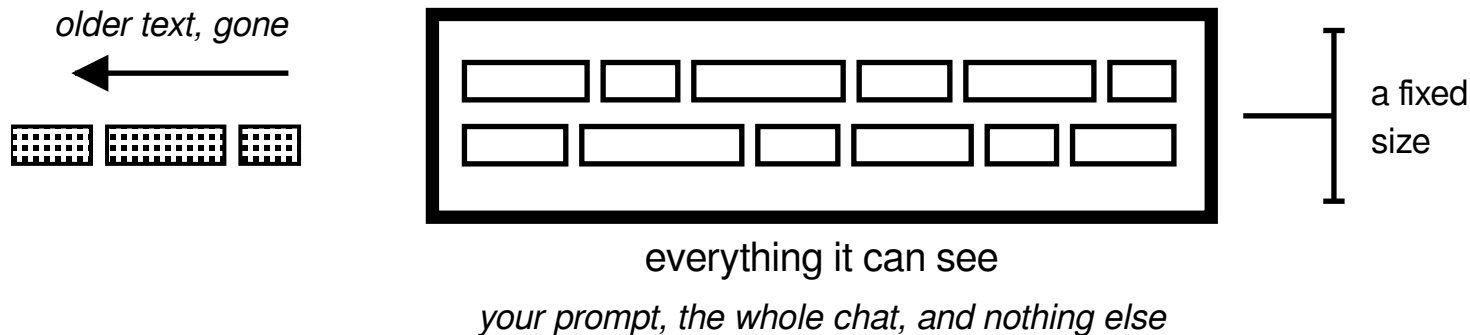
Next

Start Over

The context is everything it knows

Nothing carries over between conversations. Whatever it seems to remember about you was sent again, in the text.

The window holding that text has a size, and the oldest of it falls out.



Back

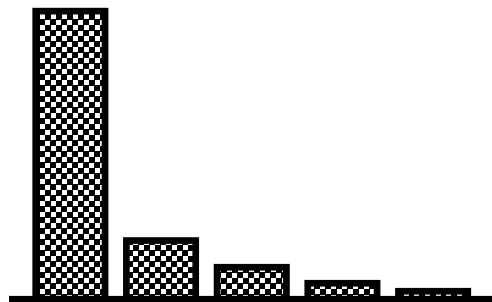
Next

Start Over

Temperature is how boldly it picks

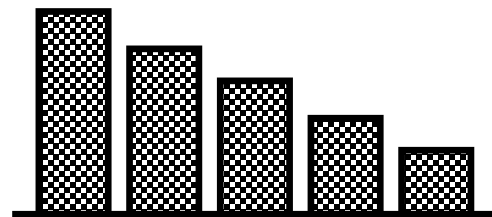
The scores do not change. What changes is how far down the list it is willing to reach.

Low repeats itself. High invents, and is wrong more often. Neither one is a truth setting.



temperature 0.2

the same answer every time



temperature 1.2

surprising, and wrong more often

Back

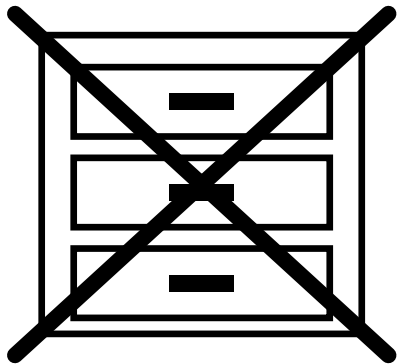
Next

Start Over

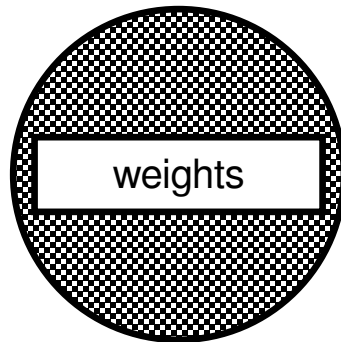
It is not looking anything up

No index, no database, no search. The answer is computed from a great many numbers, so it reads exactly as fluently when it is wrong as when it is right.

If it has to be right about something, put the something in the prompt.



nothing is looked up



every answer is computed

Back

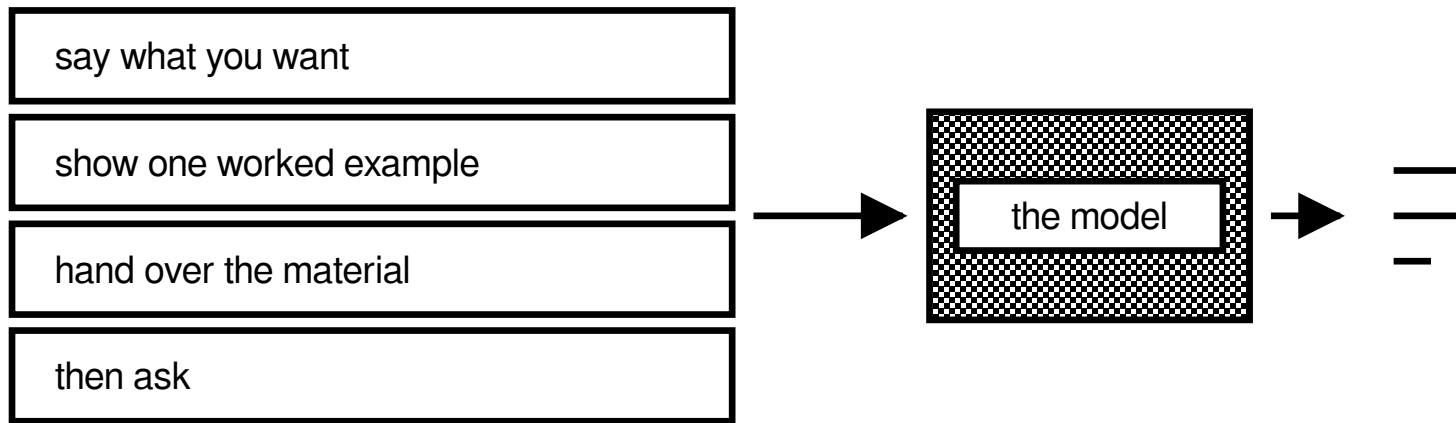
Next

Start Over

Prompting is just more context

There are no magic words. Telling it to be accurate does nothing; showing it one worked example does a great deal.

Say what you want, show it once, hand over the material, then ask.



there are no magic words — this is the whole trick

Back

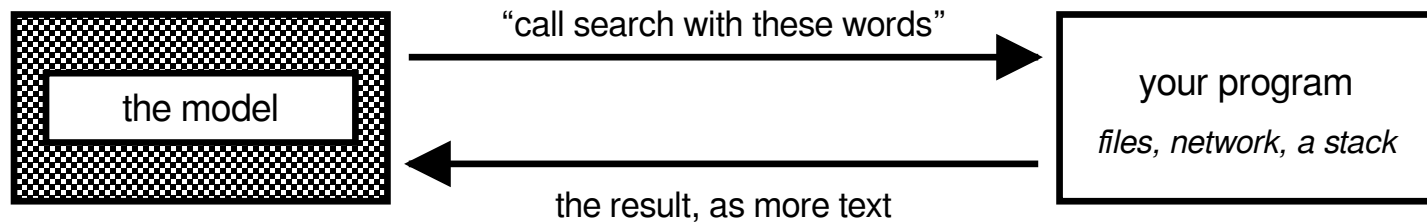
Next

Start Over

Tools: it asks, something else does

A model can only emit text. It cannot open a file or call anything. So it emits a request, a program outside it does the work, and the result comes back as more text.

HyperLab's assistant works this way, which is why you can undo what it does.



*round and round until it has an answer
— that is all an agent is*

Back

Next

Start Over

Now ask a real one

Everything above was written in advance. This is not. Set a model up under AI ? Show Assistant ? Settings, then press Ask.

With none set up it says so and the stack keeps working, which is the rule the runtime enforces.

Explain a context window to someone who has never heard of one.

Ask

Back

Next

Start Over